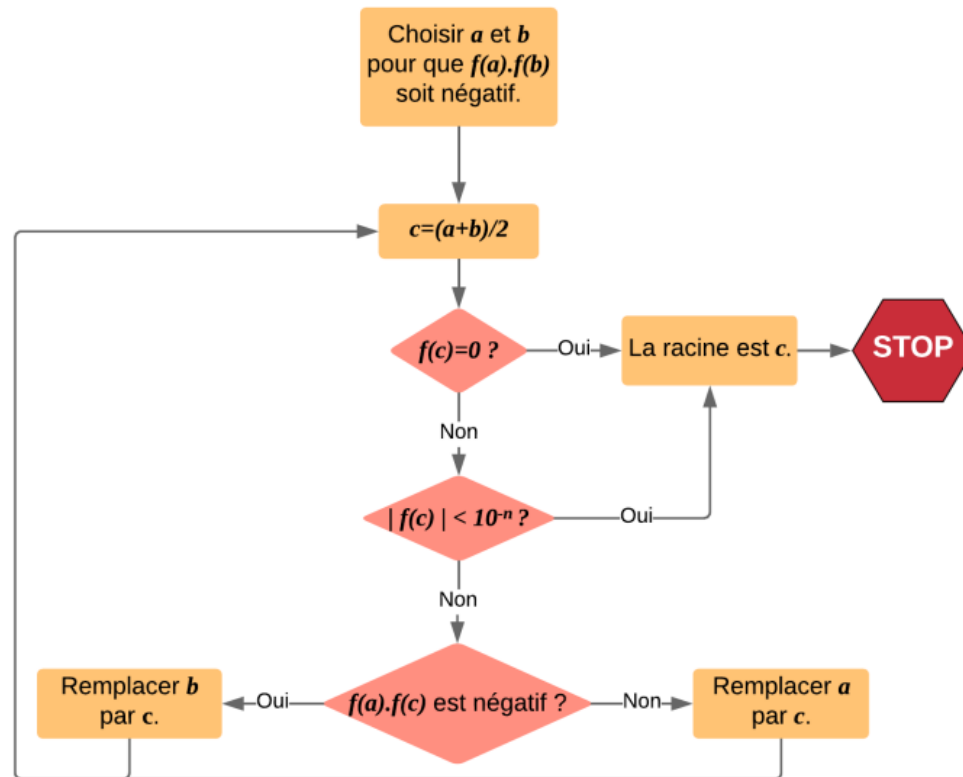


Méthode de Bisection



$$f(x) = x^2 - 4$$

$$a \leftarrow 1 \quad f(a) = -3$$

$$b \leftarrow 4 \quad f(b) = 12$$

$$c \leftarrow \frac{1+4}{2} = 2.5$$

$$f(c) = 2.25$$

$$a \leftarrow 1$$

$$b \leftarrow 2.5$$

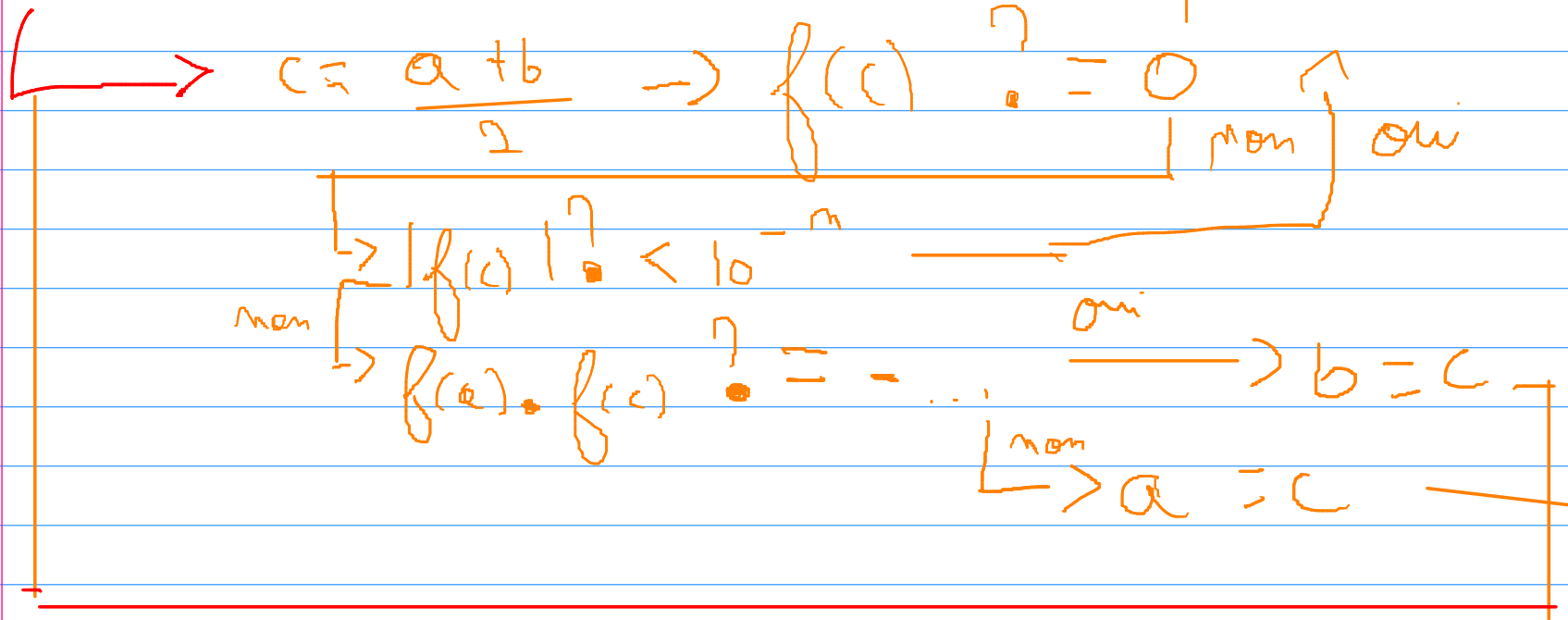
$$x^2 - 4$$

$$a = 1$$

$$\leftarrow f(a) \cdot f(b) = - \dots$$

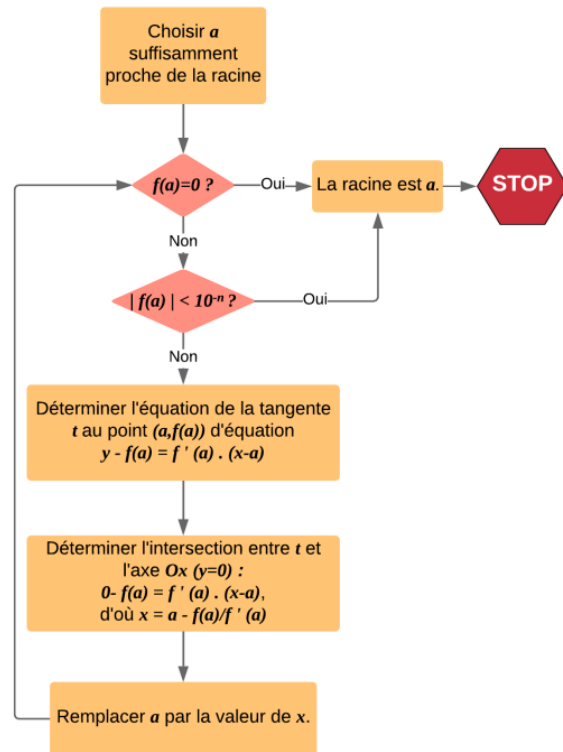
$$b = 4$$

oui
 $\rightarrow C = \text{zéro}$ stop



ex: $x^2 - 4 \rightarrow a = 1 \quad 4 \rightarrow 1,75 \quad 1,75$ zéro ≈ 2
 $b = 4 \rightarrow 2,5 \quad 2,5 \rightarrow 2,125$

Méthode de Newton



$$f(x) = x^2 - 4$$
$$f'(x) = 2x$$

$$a = 5.5$$

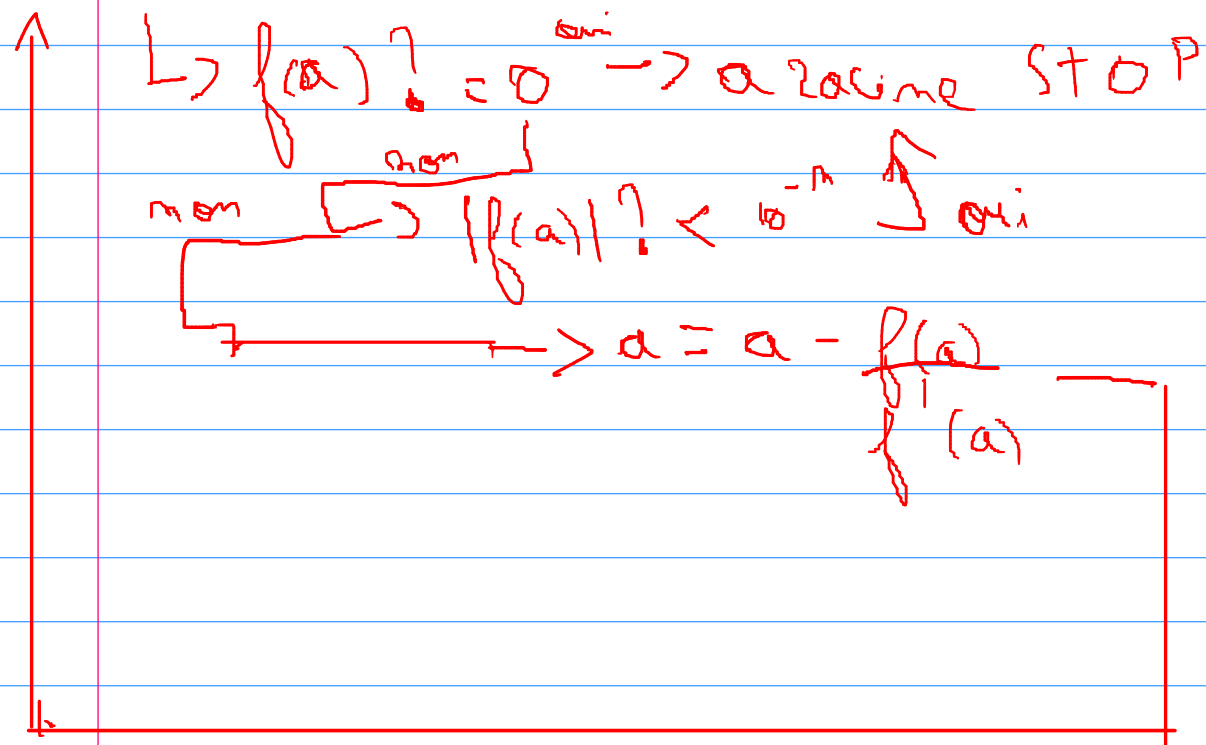
$$f(a) = 26.25$$

$$f'(a) = 11$$

$$a \leftarrow 5.5 - \frac{26.25}{11} = 3.114 \dots$$

$$x^2 - 4 \quad f'(x) = 2x$$

$a = 5,5 \leftarrow$ a good value



$$a = 5,5 \rightarrow 3,114 \rightarrow 2,199 \rightarrow 2,009$$

b = bisection

esco N = newton

$$\begin{matrix} -1 & 1 \\ \uparrow & \uparrow \\ 1 & 1 \end{matrix}$$

b a) $x^3 + x - 1$

$a=0 \quad b=1$

$f(a) \cdot f(b) = -1$

$c = \frac{a+b}{2} = \frac{1}{2} = 0,5$

$f(c) = -0,375$

$f(a) \cdot f(c) = 0,375$

$a=0,5 \quad b=1$

$c=0,75$

$f(c) = 0,171875$

$f(a) \cdot f(c) = - \dots$

$a=0,5 \quad b=0,75$

$c=0,625$

$f(c) = -0,1309$

racines entre 0,625 et 0,75

N a) $x^3 + x - 1 \quad a=0$

$f'(x) = 3x^2 + 1$

$f(a) = -1 \quad f'(a) = 1$

$a = 0 - \frac{(-1)}{1} = 1$

$f(a) = -1 \quad f'(a) = 4$

$a = 1 - \frac{-1}{4} = \frac{3}{4}$

$f(a) = \frac{11}{64}$

$f'(a) = \frac{43}{16}$

$\frac{3}{4} - \frac{\frac{11}{64}}{\frac{43}{16}} = 0,00894$